

Cristian Ramirez Rodriguez

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Research Interests

My research interests include experimental atomic and optical physics, as well as interdisciplinary work around STEM education. Currently, I am developing an evaporative cooling system to achieve nano-Kelvin temperatures under the guidance of Dr. Brynle Barrett.

Education

University of New Brunswick, Fredericton, NB, Canada

M.S., Physics

Expected May 2026

Saint Mary's College of California, Moraga, CA

B.S., Physics, *Summa Cum Laude*

GPA: 3.96

May 2023

Relevant Coursework: Analytical Mechanics, Astrophysics, Computational Physics, Differential Equations, Electricity & Magnetism, Electronics & Instrumentation, Linear Algebra, Modern Physics, Optics, Power Electronics, STEM Argument and Research, Thermal & Statistical Physics, Quantum Mechanics

Honors and Grants

Physics Departmental Scholarship (2019 - 2023)

Presidential Scholarship (2019 - 2023)

Presentations

SMC Student Research Conference & Showcase, Saint Mary's College of California, Moraga, CA, May 2024. Ramirez Rodriguez, C. "[Lagrangian Motion on Rotating Monomial Wires](#)" (poster).

SMC Student Research Conference & Showcase, Saint Mary's College of California, Moraga, CA, May 2023. Ramirez Rodriguez, C; Jerome, J; Matsuura, J; Lee, A. "[Calculating the Mass of Jupiter Through Galilean Observations](#)" (poster)

Preprints & Publications

Academic:

"A Naive Model of Covid-19 Spread From A Dry Cough" *arXiv*. 2024, doi.org/10.48550/ARXIV.2403.06435

Creative:

"[Love In The Time of Social Stratification](#)." *Passengers Journal*, June 2024

"[U](#)." *Fatal Flaw Literary Magazine*, May 2024

"[If You Find Me](#)." *¡Pa'lante!*, Cerritos College, no. 5, 2024, p. 68

"[¿Que Es Santo?](#)" *Meetinghouse*, Dartmouth College, no. 3, 2023, pp. 78–80.

"[m\ddot{y} + \(\omega^2\)y = 0](#)." *EcoTheo Review*, EcoTheo Collective, October 2022

Awards

Best Presentation in Sciences (Saint Mary's College of California - Student Research Conference, 2024)

Best Group Presentation (Saint Mary's College of California - Student Research Conference, 2023)

[Lorca Latinx Poetry Prize](#), Runner-up, 2022

Work Experience

Apr 2024 - May 2024 Lafayette Academy, AP Physics and Mathematics Instructor, Lafayette, CA

- Prepared 7 students weekly in one-on-one sessions for their exams.
- Jul 2023 - Mar 2024 East Bay Innovation Academy, Upper School Teacher, Oakland, CA
- Sole instructor for 125 High School Students in Algebra 1 and Geometry, as well as substitute for physics and computer science.
 - Developed a data-driven math curriculum incorporating several educational technologies and industry tools such as Python and Excel.
- Jun 2023 - Sep 2023 Economics Applications, Inc., Physics Consultant, Novato, CA
- Interned as Technical Lead and Project Manager for a project to determine the feasibility of deploying data centers in emerging markets.
 - Synthesized, analyzed and interpreted complex data sets to produce technical reports used by customers to attract investors.
- Sep 2022 - May 2023 Saint Mary's College of California, Teaching Assistant for Dr. Andrew Conner and Dr. Fazal Abbas in Math for Scientists, Moraga, CA
- Taught a remedial algebra course geared towards first-year undergraduate STEM majors focused on introducing concepts in an interdisciplinary way for 4 sections of 10 students.
- Sep 2020 - May 2023 Saint Mary's College of California, Stem Center Tutor, Moraga, CA
- Completed 1000+ hour tutoring 200+ students in upper division Physics, Math, Computer Science, and Data Science courses.
- Sep 2020 - Dec 2021 Saint Mary's College of California, Teaching Assistant for Dr. Ellen Veomett and Dr. Kristen Beck in Calculus 1, Moraga, CA
- Taught a virtual intro Calculus course for 8 sections of 10 students.

Leadership/Service

- Sep 2020 - May 2023 President of La Hermandad at Saint Mary's College of California
- Led a seven-member cross-functional team orchestrating cultural outreach, organizing two of the largest events on campus, the Guadalupe celebrations with over 500 attendees and the Día Del Niño celebration with over 200 attendees and engaging the local community in high-impact activities: food bank volunteering, bone marrow drives, and educational panels leading to me being named the [Latinx Graduation](#) keynote speaker.
- Jan 2021 - June 2021 Volunteer Staff, Oakland International School,
- Tutored Algebra 1 to high-school students bilingually in English/Spanish

Certifications

Social and Behavioral Responsible Conduct of Research, *CITI Program* - July 2023
 Credential ID 38633487

Google Data Analytics Specialization, *Coursera* - January 2023
 Credential ID QJCKF7BYGBFH

Data Analysis with R Programming, *Coursera* - December 2022
 Credential ID 9M426A96E9ND